

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$S = \int d^2x \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi + \frac{1}{2} \lambda_{\mu\nu} (\partial^\mu \phi - \epsilon^{\mu\sigma} \partial_\sigma \phi) (\partial^\nu \phi - \epsilon^{\nu\rho} \partial_\rho \phi) \right],$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\text{R}_\text{D}(\lambda, \delta) = \langle \lambda, \delta \rangle + \sum_{1 \leq \ell \leq 2} \frac{V_\ell / V_{\ell, XX}}{4\lambda_\ell} + \frac{1}{4} V_o^\top \underbrace{\left(\lambda_1 V_{1, XX}^{-1} + \lambda_2 V_{2, XX}^{-1} \right)}_{=\Lambda_\lambda}^{-1} V_o$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\frac{\partial V}{\partial \Lambda} = \frac{1}{\Lambda^2} \frac{\partial V}{\partial \psi_{\alpha a}} C'(0) \frac{\partial V}{\partial \bar{\psi}_a^\alpha} - \tau \Lambda^{D-2} \frac{\partial^2 V}{\partial \psi_{\alpha a} \partial \bar{\psi}_a^\alpha},$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$f_3(r) = 2\sqrt{\frac{\bar{s}^2 - Qc}{Qc}} e^{-\tilde{c}\sqrt{\bar{s}^2 - Qc}} \frac{\left(1 - \frac{r_0^2}{r^2}\right)^{(1 + \sqrt{(\bar{s}^2 - Qc)/D_2})/2}}{1 - e^{-2\tilde{c}\sqrt{\bar{s}^2 - Qc}} \left(1 - \frac{r_0^2}{r^2}\right)^{\sqrt{(\bar{s}^2 - Qc)/D_2}}}.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\tilde{H} = \frac{\sqrt{\alpha'}}{2} g^{\mu\nu} (P_\mu - \tilde{Q}_E \mathcal{A}_\mu - \tilde{Q}_M \tilde{\mathcal{A}}_\mu) (P_\nu - \tilde{Q}_E \mathcal{A}_\nu - \tilde{Q}_M \tilde{\mathcal{A}}_\nu) + \frac{\sqrt{\alpha'}}{2k^2} (\tilde{Q}_E^2 + \tilde{Q}_M^2),$$